

DUCATI MONDANI

Systems Engineer | Mechanical Design | Robotics & Automation

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PROFESSIONAL SUMMARY

Mechanical design-focused Systems Engineer with a B.S. in Mechanical Engineering and hands-on experience developing complex electromechanical and robotic systems from concept through fabrication, integration, and commissioning. Strong background in SolidWorks, mechanical and electrical design, industrial automation, robotics, PLC/SCADA systems, troubleshooting, and vendor coordination. Experienced working independently across mechanical, electrical, controls, software, and manufacturing disciplines.

CORE SKILLS

Mechanical Design	SolidWorks (CSWA), CAD, Sheet Metal, DFM, GD&T, SolidWorks Simulation, Solid Edge
Robotics & Automation	Omron Robotics, Universal Robots, Industrial Automation, System Integration, Commissioning
Electrical & Controls	Allen-Bradley / Studio 5000, Siemens STEP 7, PLC Ladder Logic, Ignition SCADA (Core Certified), EPLAN, EtherNet/IP, Modbus
Software & Analysis	Python, SQL, PostgreSQL, MATLAB, Technical Documentation, Troubleshooting, Project Management

PROFESSIONAL EXPERIENCE

Systems Engineer — Calcon Systems

Feb 2025–Present

- Design and integrate custom automation systems involving mechanical assemblies, electrical controls, PLCs, robotics, SCADA, and industrial communications.
- Serve as primary engineer for a mobile collaborative robotic platform, leading mechanical design in SolidWorks, electrical design, controls/software integration, fabrication support, vendor coordination, and final commissioning.
- Designed the complete custom platform above an autonomous mobile robot, including sheet-metal, machined, and welded components and an integrated electrical enclosure.
- Develop and modify Ignition SCADA applications for machine control, data acquisition, configurable test/task setup, visualization, and reporting using Python, SQL, and PostgreSQL.
- Program and integrate Omron and Universal Robots with Allen-Bradley PLC-controlled equipment for automated operation.
- Provide independent on-call field support, troubleshooting and modernizing legacy and current industrial systems including PLCs, VFDs, servos, safety circuits, sensors, industrial networks, and 24 VDC–480 VAC equipment.

Project Engineer, Mechanical & Electrical Systems —

Jun 2022–Dec 2024

Process Metrix / Vesuvius

Pleasanton, CA

- Managed engineering projects from design through commissioning and Site Acceptance Testing (SAT) for million-dollar industrial laser systems.
- Designed mechanical components for manufacturability, serviceability, and field installation, including CAD updates for lifting, access, and simplified machining.
- Migrated a custom PCB-based I/O system to an off-the-shelf PLC architecture, reducing system cost by 20.42%.
- Developed standardized PLC logic for Modbus communication with Windows-based software and supported robotic, machine-vision, and laser-based automation projects.
- Created detailed electrical schematics and control-system designs in EPLAN and validated positioning algorithms using 3D cameras and laser scanners.

ADDITIONAL ENGINEERING & MANUFACTURING EXPERIENCE

Driver Controls & Rear Suspension Lead — Nevada Electric

Aug 2020–May 2022

Racing — Formula SAE

University of Nevada, Reno

- Led junior engineers in development of rear suspension and driver-control systems for an electric Formula SAE race car.
- Used SolidWorks Simulation for structural analysis of suspension components and thermal analysis of brake components.
- Collaborated on MATLAB-based suspension optimization and coordinated design work across multidisciplinary engineering teams.

Manufacturing Support Specialist — Vesuvius USA

Jan 2017–Aug 2019

- Supported manufacturing and integration of complex electromechanical laser systems, troubleshooting mechanical and electrical issues with field technicians.
- Developed electrical diagrams, assembly documentation, and field installation instructions for new and one-off equipment.
- Managed procurement and supply-chain activities for high-value production systems while coordinating with engineering, manufacturing, and vendors.

Manufacturing Intern — Inverse Solutions Inc.

Jun 2014–Aug 2015

- Inspected and assembled precision CNC-machined components using technical drawings, GD&T principles, and precision measurement equipment.
- Performed quality-control inspections, deburring, and subassembly work to prepare components for production integration.

Lead CAD Designer & Club President — FIRST Robotics

Jan 2012–Apr 2015

Competition

- Led a 70+ student robotics team and designed complete competition robot assemblies in SolidWorks.
- Trained team members in CAD and mechanical design; received a FIRST Excellence in Engineering Award for a unique climbing mechanism.

EDUCATION

University of Nevada, Reno

May 2022

B.S. Mechanical Engineering | Minors: Manufacturing Quality and Mathematics

CERTIFICATIONS

Inductive Automation Ignition Core Certification

Solidworks CSWA